CS 301 · MEETING 12 · CLOSING

Back to the O

Can we reduce the number of operations?

The exponent can change

Algorithm or bound	Arithmetic exponent
The three-loop algorithm	3
Strassen (1969)	$\log_2 7 \approx 2.807$
Dupont et al. (August 2026)	$\omega < 2.371177$
OpenAI preprint (October 2026)	**$\omega \leq 9/4 = 2.25$**

- Our cubic count belongs to a particular algorithm
- OpenAI reports a lower upper bound on the exponent

Paper: October 2, 2026 · OpenAI release: October 6, 2026

The exponent and the seconds

$$O\left(n^{9/4+\varepsilon}\right) \quad \text{for every fixed } \varepsilon > 0$$

Arithmetic operations for square complex matrices

- A smaller exponent reduces growth with matrix size
- Constants, memory traffic, and precision still matter
- The preprint gives no competitive finite matrix size

OpenAI, An Upper Bound of 9/4 for the Matrix Multiplication Exponent (2026)